

Rahul Ghoshal

Panchla, Howrah, India

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Professional Summary

Data Science postgraduate with strong foundations in machine learning, statistics, and data analytics. Hands-on experience in predictive modeling, NLP-based sentiment analysis, and data visualization using Python, SQL, R, and Power BI, with internship experience at ISI Kolkata.

Education

Master of Science in Data Science

2024 – 2026

JIS University, Agarpara, Kolkata

CGPA: **9.57** / **10.0**[First Year]

Bachelor of Science in Computer Science (Honours)

2021 – 2024

Vivekananda College, Madhyamgram, Kolkata

CGPA: **8.97** / **10.0**

Technical Skills

- **Data Analysis/Spreadsheet Tools:** Microsoft Excel (Advanced Excel, Pivot Tables, Power Query, XLOOKUP, INDEX-MATCH)
- **Programming Languages:** Python, R, SQL.
- **Libraries & Frameworks:** Pandas, NumPy, Scikit-learn.
- **Databases:** MySQL.
- **Data Visualization:** Power BI, Matplotlib, Seaborn.
- **Machine Learning:** Regression, Classification, Feature Engineering, Exploratory Data Analysis.

Internship Experience

Data Science Intern

May 2025 – July 2025

IDEAS TIH, Indian Statistical Institute (ISI), Kolkata

- Collected, cleaned, and analyzed multi-platform customer reviews (Website, Instagram, Facebook) from FY 2022–2025 to build an end-to-end sentiment analysis pipeline.
- Implemented VADER-based NLP models with custom compound score logic to classify feedback into five sentiment categories.
- Detected a **27.2%** drop in "Highly Positive" sentiment (**81.6%** → **54.4%**) in FY2023–24.
- Performed year-wise and platform-wise trend analysis to identify key customer satisfaction patterns.
- Developed data visualizations and insight dashboards using Matplotlib and Seaborn to support stakeholder decision-making.

Projects

Customer Behavior Analysis (GitHub Link)

- Analyzed a customer purchase dataset using Python (Pandas) to perform data cleaning, preprocessing, and Exploratory Data Analysis (EDA).
- Executed analytical SQL queries in MySQL Server to identify customer segments, purchasing patterns, and discount impact on sales.
- Developed an interactive Power BI dashboard to visualize customer segmentation, purchase trends, and key business insights.

Hypertension Prediction System (GitHub Link)

- Built a Random Forest-based hypertension prediction system on 1,980+ patient records across 3 classes (Normal, Prehypertension, Hypertension), achieving **97.4% classification accuracy**.
- Processed and engineered clinical data by performing cleaning, feature encoding, EDA, and correlation analysis using Pandas and NumPy.
- Developed an interactive Streamlit web interface enabling real-time risk prediction from multiple clinical parameters.